M-Tech Production & Marketing

Technology | Training | Marketing | Projects

Week 3 - Day 4 Report

Internship Program — Batch 2026

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1. Introduction

This report presents the work completed for Experiment 12: Generative AI – GANs, VAEs & Diffusion Models under the M-Tech Production & Marketing AI/ML Internship Program 2026. The experiment focused on understanding the rapidly growing field of Generative Artificial Intelligence, which enables machines to create new content such as images, text, audio, and videos. Different generative models were studied to understand their architecture, functionality, and practical applications

2. Objectives

- Understand the concept of Generative Artificial Intelligence.
- Learn the working principles of GANs, VAEs, and Diffusion Models.
- Compare different generative learning techniques.
- Explore real-world applications of AI-generated content.
- Understand the advantages and limitations of modern generative models.

3. Task Description

The experiment introduced three important Generative AI models: Generative Adversarial Networks (GANs), Variational Autoencoders (VAEs), and Diffusion Models. It explained how each model learns from existing datasets to generate realistic and meaningful content. The experiment also discussed practical use cases of these models in image generation, creative design, healthcare, and entertainment.

4. Work Done

During this experiment, I explored the basic concepts of Generative AI and learned how different models generate new data instead of simply making predictions. I studied the working mechanism of GANs, where a Generator creates synthetic data while a Discriminator evaluates its quality until realistic results are achieved. I also learned how Variational Autoencoders compress and reconstruct data to generate new samples with similar characteristics. Furthermore, I studied Diffusion Models, which gradually remove noise from random data to produce highly realistic images. Through examples and demonstrations, I gained an understanding of how these technologies are used in AI image generation, content creation, medical imaging, and multimedia applications

5. Challenges Faced

• Understanding the training process of GANs.

• Comparing the architecture of GANs, VAEs, and Diffusion Models.

• Learning how diffusion-based image generation produces realistic outputs..

6. Key Learnings

• Generative AI focuses on creating new content from learned data patterns.

• GANs use Generator and Discriminator networks to improve generated results.

• VAEs create compressed data representations for content generation.

• Diffusion Models generate high-quality images through gradual denoising.

• Generative AI has wide applications in design, healthcare, entertainment, and digital media.

7. Conclusion

This experiment expanded my knowledge of modern Generative AI technologies and their practical importance. I gained a better understanding of how GANs, VAEs, and Diffusion Models operate and how they are transforming industries through AI-generated content. The concepts learned in this experiment will support my future work in advanced Artificial Intelligence and Machine Learning.

Intern Signature:	____________	**Supervisor:**	
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